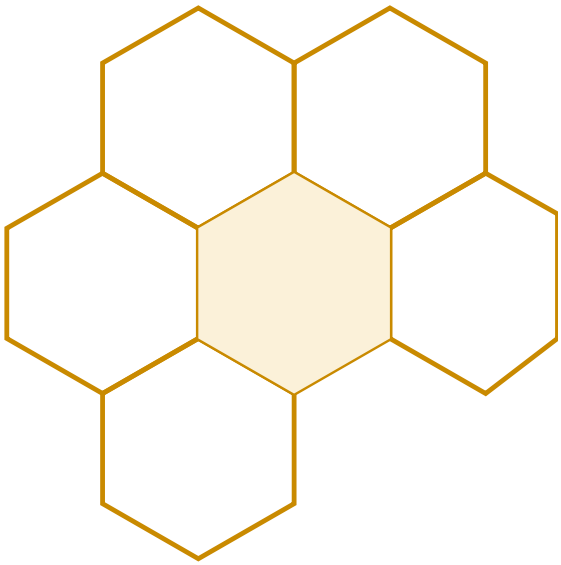
EUROAVIA IDEATHON 2026 · ANSYS CHALLENGE



Material catalogue

31 aerospace materials with sourced, tagged properties and index rankings

3 November 2026 · generated from the material pool

Team Beetiful

Generated by `python scripts/material_catalogue.py` from `aerokit.materials.pool` (data: `research_notes/Malzeme havuzu/materials_pool.json`, 31 materials, every value with its source and a tag: D read from the primary source, S secondary, Y estimate or stand-in). Strength columns mix MIL-HDBK-5J A/S-basis allowables and datasheet typical values; the source string in the pool says which. Fatigue is the stated condition of each source (mostly 10^7 cycles, R = −1). Do not edit this file by hand: change the pool and regenerate.

Aluminium alloys

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
Al 2024-T3 sheet	2768 D	72.4 D	324 D	441 D	124 Y
Al 2024-T351 plate	2768 D	73.8 D	331 D	441 D	160 Y
Al-Li 2050-T84 plate	2700 S	76.5 S	476 S	503 S	–
Al-Li 2198-T8 sheet	2700 S	76.5 S	407 S	462 S	–
Al 6061-T6 sheet	2713 D	68.3 D	248 D	290 D	109 Y
Al 7050-T7451 plate	2823 D	71.0 D	441 D	510 D	100 Y
Al 7075-T6 sheet	2796 D	71.0 D	483 D	538 D	170 Y
Al 7075-T73 / T7351	2796 D	71.0 D	393 D	469 D	–
Al 2618-T61 forging	2768 D	73.8 D	324 D	400 D	–
AlSi10Mg L-PBF	2670 D	72.0 S	220 D	340 D	110 D
Scalmalloy L-PBF	2670 D	69.0 S	490 D	520 D	87 D
A20X / A205 L-PBF	2850 D	79.0 D	390 D	450 D	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
Al 2024-T3 sheet	119.0 D	22.9 D	120 Y	12.8 D	sheet forming, machined, riveted assembly
Al 2024-T351 plate	119.0 D	22.9 D	120 Y	12.8 D	machined from plate
Al-Li 2050-T84 plate	–	–	–	12.8 Y	machined from thick plate (12-165 mm)
Al-Li 2198-T8 sheet	–	–	–	12.8 Y	sheet forming (fuselage skin)
Al 6061-T6 sheet	154.0 D	22.9 D	–	12.8 D	sheet, extrusion, machined, weldable
Al 7050-T7451 plate	157.5 D	23.0 D	–	12.8 D	machined from thick plate
Al 7075-T6 sheet	131.0 D	22.9 D	100 Y	12.8 D	sheet, machined from T651 plate
Al 7075-T73 / T7351	156.0 D	22.9 D	–	12.8 D	sheet (T73), machined from plate (T7351), forgings
Al 2618-T61 forging	156.0 D	20.8 D	316 D	12.8 D	die/hand forging, machined
AlSi10Mg L-PBF	110.0 D	20.0 D	–	12.0 S	AM L-PBF, machined critical faces, stress relief 270 C/90 mi
Scalmalloy L-PBF	–	–	–	128.0 Y	AM L-PBF, aged 325 C/4 h, machined critical faces
A20X / A205 L-PBF	–	–	200 Y	12.0 Y	AM L-PBF, proprietary T7 heat treatment, casting (A20X)

Titanium alloys

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
Ti-6Al-4V annealed (wrought)	4429 D	110.3 D	869 D	924 D	331 D
Ti-6Al-4V L-PBF	4410 D	110.0 D	945 D	1055 D	600 S
Ti-5553 STA	4640 D	–	1000 D	1050 D	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
Ti-6Al-4V annealed (wrought)	7.3 D	8.8 D	–	46.5 S	machined from plate/bar, forging
Ti-6Al-4V L-PBF	7.3 Y	8.8 Y	350 D	46.5 S	AM L-PBF, stress relief 800 C/2 h, HIP, machined critical fa
Ti-5553 STA	–	–	–	46.5 S	forging (landing gear, structural), machined

Steels

Mechanical:

Material	ρ (kg/m^3)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
17-4PH H1025	7806 D	196.5 D	1000 D	1069 D	–
15-5PH H1025	7833 D	196.5 D	1000 D	1069 D	1052 Y
300M (0.42C, AMS 6257/6419)	7833 D	199.9 D	1586 D	1931 D	634 Y

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
17-4PH H1025	17.3 D	10.8 Y	316 S	5.0 S	machined from bar, AM L-PBF (17-4PH powder)
15-5PH H1025	17.8 D	11.0 Y	–	5.0 S	machined from bar/plate, forging
300M (0.42C, AMS 6257/6419)	35.0 Y	12.2 Y	–	2.8 Y	forging (landing gear), machined, quench and temper

Nickel alloys

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
Inconel 718 wrought (STA)	8221 D	202.7 D	1034 D	1276 D	298 Y
Inconel 718 L-PBF	8150 D	200.0 Y	1145 D	1375 D	–
Inconel 625 L-PBF	8400 D	204.0 D	640 D	890 D	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
Inconel 718 wrought (STA)	11.4 D	13.2 D	704 D	11.6 S	machined from bar/forging, sheet
Inconel 718 L-PBF	–	13.1 D	700 D	11.6 Y	AM L-PBF, solution + double age (AMS 2774/5662), HIP optiona
Inconel 625 L-PBF	–	–	–	11.6 Y	AM L-PBF, stress relief 870 C/1 h

Magnesium alloys

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
Mg AZ31B-H24 sheet	1769 D	44.8 D	200 D	269 D	66 Y
Mg WE43B / Elektron 43	1830 S	45.0 S	172 S	–	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
Mg AZ31B-H24 sheet	76.0 D	25.9 D	–	25.8 D	sheet (warm forming), extrusion, machined
Mg WE43B / Elektron 43	51.0 S	–	250 S	25.8 D	casting (WE43B T6), wrought plate/extrusion/forging (Elektro

Composites

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
CFRP QI IM7/8552	1570 D	57.8 D	–	722 D	–
GFRP 7781 E-glass / epoxy (913)	1830 D	22.0 D	–	450 D	–
CF/PEEK (Toray Cetex TC1200)	1530 D	40.8 D	–	578 D	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
CFRP QI IM7/8552	2.7 Y	2.4 Y	121 D	23.4 Y	autoclave prepreg [25/50/25], AFP
GFRP 7781 E-glass / epoxy (913)	–	–	–	8.1 D	autoclave/oven prepreg (8HS fabric)
CF/PEEK (Toray Cetex TC1200)	–	–	–	23.3 Y	thermoplastic press forming, thermoforming, welding

Polymers

Mechanical:

Material	ρ (kg/m³)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
PEEK 450G (neat)	1300 D	4.0 D	98 D	–	–
ULTEM 9085 (FDM)	1270 D	2.5 D	70 D	68 D	–
PA12 SLS (EOS PA 2200)	930 D	1.6 D	–	48 D	–
PA12-CF (Stratasys FDM Nylon 12CF)	1190 D	9.5 D	–	84 D	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
PEEK 450G (neat)	0.3 D	55.0 D	240 D	–	injection moulding, extrusion, machined stock
ULTEM 9085 (FDM)	–	45.0 D	173 D	–	FDM (Fortus 450mc / F900)
PA12 SLS (EOS PA 2200)	0.1 D	109.0 D	80 D	9.1 Y	SLS
PA12-CF (Stratasys FDM Nylon 12CF)	0.6 D	–	130 D	–	FDM (Fortus 450mc, F900, F3300)

Sandwich cores

Mechanical:

Material	ρ (kg/m^3)	E (GPa)	Fty (MPa)	Ftu (MPa)	Fatigue (MPa)
Nomex honeycomb HRH-10 1/8-3.0	48 D	–	–	–	–

Thermal, carbon and routes:

Material	k (W/m K)	α (10^{-6}/K)	Tmax (°C)	CO_2 (kg/kg)	Routes
Nomex honeycomb HRH-10 1/8-3.0	–	–	–	–	sandwich core (bonded faces)

Performance-index rankings over the pool

Relative mass of a part of fixed geometry family and fixed stiffness or strength (1.00 = lightest). Materials without the property are left out (composites and polymers have no Fty entry).

- **stiff tie E/ρ**: CFRP QI IM7/8552 1.00 · Al-Li 2050-T84 plate 1.30 · Al-Li 2198-T8 sheet 1.30 · A20X / A205 L-PBF 1.33 · AlSi10Mg L-PBF 1.37 · CF/PEEK (Toray Cetex TC1200) 1.38
- **stiff beam E^½/ρ**: CFRP QI IM7/8552 1.00 · CF/PEEK (Toray Cetex TC1200) 1.16 · Mg AZ31B-H24 sheet 1.28 · Mg WE43B / Elektron 43 1.32 · Al-Li 2050-T84 plate 1.49 · Al-Li 2198-T8 sheet 1.49
- **stiff panel E^⅓/ρ**: CFRP QI IM7/8552 1.00 · CF/PEEK (Toray Cetex TC1200) 1.09 · Mg AZ31B-H24 sheet 1.23 · Mg WE43B / Elektron 43 1.27 · PA12-CF (Stratasys FDM Nylon 12CF) 1.39 · Al-Li 2050-T84 plate 1.57
- **strong tie σ/ρ**: Ti-5553 STA 1.00 · Ti-6Al-4V L-PBF 1.01 · 300M (0.42C, AMS 6257/6419) 1.06 · Ti-6Al-4V annealed (wrought) 1.10 · Scalmalloy L-PBF 1.17 · Al-Li 2050-T84 plate 1.22
- **strong beam σ^⅔/ρ**: Scalmalloy L-PBF 1.00 · Al-Li 2050-T84 plate 1.03 · Al 7075-T6 sheet 1.06 · Ti-6Al-4V L-PBF 1.07 · Ti-5553 STA 1.08 · Ti-6Al-4V annealed (wrought) 1.13
- **strong panel σ^½/ρ**: Scalmalloy L-PBF 1.00 · Al-Li 2050-T84 plate 1.03 · Mg AZ31B-H24 sheet 1.04 · Al 7075-T6 sheet 1.05 · PEEK 450G (neat) 1.09 · Al-Li 2198-T8 sheet 1.11

The same indices with embodied CO_2 instead of mass

Ranking by $\rho \cdot CO_2$ in place of ρ: the part with the least production CO_2. For a part that flies, the use phase is about 98 % of life CO_2 (docs 11–13), so this ranking matters only when there is no use phase (ground equipment, tooling, a satellite judged on production).

- **stiff beam $E^{1/2}/\rho$, CO_2**: 300M (0.42C, AMS 6257/6419) 1.00 · 17-4PH H1025 1.81 · 15-5PH H1025 1.82 · GFRP 7781 E-glass / epoxy (913) 2.06 · AlSi10Mg L-PBF 2.46 · A20X / A205 L-PBF 2.51
- **stiff panel $E^{1/3}/\rho$, CO_2**: 300M (0.42C, AMS 6257/6419) 1.00 · GFRP 7781 E-glass / epoxy (913) 1.43 · 17-4PH H1025 1.81 · 15-5PH H1025 1.82 · PA12 SLS (EOS PA 2200) 1.94 · AlSi10Mg L-PBF 2.08
- **strong beam $\sigma^{2/3}/\rho$, CO_2**: 300M (0.42C, AMS 6257/6419) 1.00 · 17-4PH H1025 2.45 · 15-5PH H1025 2.45 · Al-Li 2050-T84 plate 3.55 · Al 7075-T6 sheet 3.64 · Al 7050-T7451 plate 3.91

Flags and notes per material

- **Al 2024-T3 sheet** — fst_cabin (metal, non-combustible) [Y]; fire_resistant per 14 CFR 1.1 benchmark (Al alloy) · Design allowables (A-basis) not typical values. AA typical (aerospacemetals.com 2023, [S]) is 345/483 MPa and 2780 kg/m3 (used in build_studies.py panel study). At 150 C / 10,000 h Fty retains 82 % (MMPDS-01).
- **Al 2024-T351 plate** — fst_cabin (metal) [Y] · library.py AA typical Fty 324 MPa is BELOW this A-basis 331 MPa (known discrepancy, library.py docstring). 2024-T351 plate S-N curve not in MIL-HDBK-5J; 2024-T4 bar is the proxy.
- **Al-Li 2050-T84 plate** — fst_cabin (metal) [Y] · Constellium claims vs 7050-T7451: -4 % density, +7 % stiffness, +8 % strength, +25 % fatigue (relative, marketing). Fatigue (numeric), k, CTE, Tmax: NOT FOUND. AMS 4413.
- **Al-Li 2198-T8 sheet** — fst_cabin (metal) [Y] · Constellium brochure: '-3 % density, +3 % stiffness' vs 7050/2024 [D, relative only]. Elongation, fatigue, k, CTE, Tmax, price: NOT FOUND.
- **Al 6061-T6 sheet** — fst_cabin (metal) [Y] · AA typical (aerospacemetals.com 2023 [S]) 276/310 MPa, 2700 kg/m3; library.py fatigue_5e8 96.5 MPa (ASM/MatWeb, rotating beam). Tmax NOT FOUND.
- **Al 7050-T7451 plate** — fst_cabin (metal) [Y] · MIL A-basis 441/510 confirms the AMS 4050 snippet reading (441/510/9) in alloy_datasheets.md; distributor 'typical' 480-520 / 540-590 is not comparable. Tmax NOT FOUND (MIL Fig. 3.7.4.2.1 has the curves).
- **Al 7075-T6 sheet** — fst_cabin (metal) [Y] · AA typical (library.py, ASM/MatWeb) 503/572 MPa, 2810 kg/m3, fatigue_5e8 159 MPa: typical vs A-basis allowable, not a contradiction.
- **Al 7075-T73 / T7351** — fst_cabin (metal) [Y]; overaged: better SCC resistance than T6 · Fatigue for T73 NOT extracted (use the 7075-T6 curve only as a rough proxy). Tmax NOT FOUND.
- **Al 2618-T61 forging** — elevated-temperature Al alloy · At 200 C, 1,000 h: Fty keeps ~73 % (MMPDS-01 fig, C notes). Fatigue NOT FOUND.
- **AlSi10Mg L-PBF** — as-built surface halves fatigue strength; HIP lowers fatigue (Uzan 2017) · Tmax NOT FOUND (Cao 2021: <200 C for 30 min gives little change; EOS stress relief at 270 C). Ashby Al family CO2 tagged S here (build_studies.py tags it D; the source is a compilation).
- **Scalmalloy L-PBF** — near-isotropic XY/Z; Sc supply drives CO2 and price · Yield 319 MPa at 150 C and 60 MPa at 300 C (Colibrium sheet, via build_studies.py). k, CTE, Tmax: NOT FOUND.
- **A20X / A205 L-PBF** — MMPDS-listed A20X alloy (per Eckart sheet); AMS 4471 composition · Eckart sheet: fatigue 'similar to cast A20X and superior to A357' (no number). Eckart web page (eckart.net/en/a20x) gives 'up to' 511/440/13 for AM, 475/420/3 for A205 castings. Price, k, CTE: NOT FOUND.
- **Ti-6Al-4V annealed (wrought)** — fireproof-adjacent: 0.016 in Ti accepted as firewall without test (AC 20-135 para 6d(6)) · library.py holds Aubert & Duval minimums 827/896 MPa, 4430 kg/m3. At 300 C Fty keeps ~63 % (MMPDS-01). Tmax for wrought NOT read (EOS gives ~350 C for AM).
- **Ti-6Al-4V L-PBF** — HIP pays only with machining (Greitemeier); engine-zone candidate (fireproof benchmark) · Colibrium EBM Spectra L Gr23 as built: 907-940 YS / 986-1015 UTS (alloy_datasheets.md). As-HIPed datasheet NOT FOUND.
- **Ti-5553 STA** — quasi-beta alloy · K1c 55 MPa sqrt(m) STA / 70 BASCA; beta transus 860 C (same sheet). The extracted composition line prints 'Vanadium 2.00' (nominal 5) - layout/sheet error, composition not used. E, fatigue, k, CTE, Tmax, price: NOT FOUND (TIMET/ATI sheet not found).
- **17-4PH H1025** — fireproof benchmark material (steel) · Fatigue for H1025 NOT FOUND (MIL-HDBK-5J has an H900 bar S-N curve only, Fig. 2.6.9.1.8(a)).
- **15-5PH H1025** — fireproof benchmark material (steel) · Price, Tmax NOT FOUND.
- **300M (0.42C, AMS 6257/6419)** — needs corrosion protection / hydrogen-embrittlement care [Y] · Price, Tmax NOT FOUND.

- **Inconel 718 wrought (STA)** — fireproof candidate; engine zone · Wrought price NOT FOUND.
- **Inconel 718 L-PBF** — fireproof candidate; engine zone · Fatigue for AM 718 NOT FOUND (EOS gives no number).
- **Inconel 625 L-PBF** — corrosion/oxidation resistant · Wrought reference only: annealed 625 bar, MIL-HDBK-5J Fig. 6.3.3.1.8(a) (log Nf = 24.49 - 9.62 log Seq, Seq = Smax(1-R)^0.42) gives ~339 MPa at R=-1, 1e7, polished [Y]; no AM fatigue found. k, CTE, Tmax, price: NOT FOUND.
- **Mg AZ31B-H24 sheet** — Mg in seats prohibited by AS8049A/ETSO-C127a; allowed only via special condition (EASA SC D-32); corrosion protection required · Smiths HP minimums (O 125/220, H24 200/270 MPa) agree with MIL. Tmax and price NOT FOUND.
- **Mg WE43B / Elektron 43** — FAA flammability tested (Luxfer, snippet); seat use only via special condition (EASA SC D-32) · All values from search snippets of Luxfer PDFs that could not be opened (redirect; no bypass attempted). Smiths HP: Elektron 43 Rm 220 MPa, 1.8 [S]. Unverified secondary values (SG 1.83, E 44.1, 195/303 MPa) noted in B notes must not be cited. Ftu, elongation, fatigue, CTE, price: NOT FOUND.
- **CFRP QI IM7/8552** — epoxy: needs FR/phenolic surface for cabin OSU 65/65 [S]; EoL: continuous fibre not economically re-separable (Granta) · QI elongation, fatigue and price: NOT FOUND. Yield not applicable (linear to failure).
- **GFRP 7781 E-glass / epoxy (913)** — glass/PHENOLIC 7781 on Nomex passes OSU (HexPly 260: 44/44); epoxy 913 not demonstrated · Toray Cetex TC1100 7781/PPS: 1950 kg/m3, 21.2 GPa, 494 MPa [D]. Fatigue, k, CTE, Tmax, price: NOT FOUND.
- **CF/PEEK (Toray Cetex TC1200)** — thermoplastic: weldable/remeltable (circularity argument) [Y] · UD AS4D: 1590 kg/m3, 135 GPa, 2410 MPa. PEEK Tg 143 C limits service (Victrex 450G). FST results, fatigue, k, CTE, price: NOT FOUND.
- **PEEK 450G (neat)** — LOI 35 % at 3.2 mm, 24 % at 0.4 mm (ISO 4589); no FAR 25.853 data on sheet · CO2: only an unattributed web range 7.5-13 kg CO2e/kg (B notes) - NOT FOUND properly. Price NOT FOUND.
- **ULTEM 9085 (FDM)** — meets 14 CFR 25.853(a) 60 s and 12 s vertical burn, 15 s horizontal (Table 7); OSU heat release 14 CFR 25.853(d): peak 41.8-57.0 kW/m2, 2-min 32.7-52.8 kW min/m2 (<= 65/65); smoke BSS 7238 Ds 4-15; toxicity BSS 7239 / AITM 3.0005 reported; UL 94 V-0 (natural); EN 45545-2 tested; ASTM E595 outgassing TML 0.33-0.37 %, CVCM <=0.02 %; CG (Certified Grade) has lot traceability · Distributor sheets print other moduli (2.15-2.27 GPa ipfl; 2.85 GPa SABIC resin ISO 527). CO2 and price NOT FOUND.
- **PA12 SLS (EOS PA 2200)** — no flame retardant: 'PA2200-parts can burn' (EOS 2004); UL 94 HB only - cabin FST not demonstrated; CarbonReduced grade: '45% less CO2e' (EOS, no absolute value) · EOS 2004 sheet: 1700 +-150 MPa, 45 +-3 MPa, 20 +-5 %, 0.90-0.95 g/cm3 (older than current web values). Price NOT FOUND.
- **PA12-CF (Stratasys FDM Nylon 12CF)** — 35 wt % chopped carbon fibre; no FAR 25.853 / UL 94 data on sheet; electrically conductive · CTE in Table 3 could not be assigned to orientation from the PDF text layout (115.7 vs 37.31 um/m/C at -50 to 20 C) - omitted. This is an FDM filament, not SLS. CO2 and price NOT FOUND.
- **Nomex honeycomb HRH-10 1/8-3.0** — meets FAR 25.853 'self extinguishing' (HRH-10 PDS p3) · Shear: G_L 41 MPa, tau_L 1.21 MPa, G_W 24 MPa, tau_W 0.69 MPa (same table). Al 5052 core 1/8-3.1 (50 kg/m3) is ~7x stiffer in shear. CO2 and price NOT FOUND.